

1.1 Introduction

Any function of two or more variables may be differentiated partially with respect to one variable treating other variables as constants; for instance, the function $f(x, y)$ may be differentiated with respect to x taking y as constant and similarly we can perform partial differentiation with respect to y keeping x constant. Here partial derivatives of $f(x, y)$ with respect to x and y are denoted as $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ respectively; also the partial derivatives of $\frac{\partial f}{\partial x}$ with respect to x and y are denoted as $\frac{\partial^2 f}{\partial x^2}$ or f_{xx} and $\frac{\partial^2 f}{\partial x \partial y}$ or f_{xy} respectively.

Again, if ' $z = f(x, y)$ ' is a function of two independent variables ' x ' and ' y ', let us use the following notations for the partial derivatives of ' z ':

$$\frac{\partial z}{\partial x} \equiv p, \quad \frac{\partial z}{\partial y} \equiv q, \quad \frac{\partial^2 z}{\partial x^2} \equiv r, \quad \frac{\partial^2 z}{\partial x \partial y} \equiv s, \quad \frac{\partial^2 z}{\partial y^2} \equiv t$$

Remarks

- **Order of partial differentiation is immaterial in general, i.e.** $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$
- **Geometrical interpretation:** $\frac{\partial f}{\partial x}$ gives the slope of tangent to the curve

$z = f(x, y)$ at the point of intersection with the plane $y = 0$.

Example1 Find all the first order partial order derivatives for the function

$$t = x^3 y z + 5\sqrt{y} z + \sin \frac{1}{y} e^{x^2 y - 2y^2}$$

Solution: $\frac{\partial t}{\partial x} = 3x^2 y z + \sin \frac{1}{y} e^{x^2 y - 2y^2} (2xy)$

$$\frac{\partial t}{\partial y} = x^3 z + \frac{5z}{2\sqrt{y}} + \sin \frac{1}{y} e^{x^2 y - 2y^2} (x^2 - 4y) + \cos \frac{1}{y} \left(-\frac{1}{y^2} \right) e^{x^2 y - 2y^2}$$

$$\frac{\partial t}{\partial z} = x^3 y + 5\sqrt{y}$$

Example2 If $v = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$, show that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$

Solution: $\frac{\partial v}{\partial x} = -(x^2 + y^2 + z^2)^{-\frac{3}{2}}(2x) = -\frac{x}{(x^2+y^2+z^2)^{\frac{3}{2}}}$

$$\Rightarrow \frac{\partial^2 v}{\partial x^2} = -\frac{(x^2+y^2+z^2)^{\frac{3}{2}} - \frac{3x}{2}(x^2+y^2+z^2)^{\frac{1}{2}}(2x)}{(x^2+y^2+z^2)^3} = -\frac{(x^2+y^2+z^2)^{\frac{1}{2}}(x^2+y^2+z^2-3x^2)}{(x^2+y^2+z^2)^3} = \frac{2x^2-y^2-z^2}{(x^2+y^2+z^2)^{\frac{5}{2}}}$$

Similarly $\frac{\partial v}{\partial y} = \frac{2y^2-x^2-z^2}{(x^2+y^2+z^2)^{\frac{5}{2}}}$ and $\frac{\partial v}{\partial z} = \frac{2z^2-x^2-y^2}{(x^2+y^2+z^2)^{\frac{5}{2}}}$

$$\therefore \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = \frac{2x^2-y^2-z^2+2y^2-x^2-z^2+2z^2-x^2-y^2}{(x^2+y^2+z^2)^{\frac{5}{2}}} = 0$$

1.2 Differentiation of Homogeneous Functions

1.2.1 Homogeneous Function: A function $f(x, y)$ of two variables is said to be homogeneous function in x and y of degree n , if it can be written as $x^n \phi\left(\frac{y}{x}\right)$.

Note: The degree n of a homogeneous function may be a rational number.

* $\frac{x^2 y^2}{x^2 + y^2}$ is a homogeneous function of degree 2

$$\therefore \frac{x^2 y^2}{x^2 + y^2} = \frac{x^4 \left(\frac{y}{x}\right)^2}{x^2 \left(1 + \left(\frac{y}{x}\right)^2\right)} = x^2 \phi\left(\frac{y}{x}\right)$$

* $f(x, y) = \tan^{-1}\left(\frac{\sqrt{x} + \sqrt{y}}{x + y}\right)$ is not homogeneous function

$$\therefore \tan^{-1}\left(\frac{\sqrt{x} + \sqrt{y}}{x + y}\right) = \tan^{-1}\left(\left(\frac{\sqrt{x}}{x}\right) \frac{1 + \sqrt{\frac{y}{x}}}{1 + \frac{y}{x}}\right), \text{ which cannot be written in the form } x^n \phi\left(\frac{y}{x}\right)$$

1.2.2 Euler's Theorem on Homogeneous Functions

Let $t = f(x, y)$ be a homogeneous function in x and y of degree n ,

then $x \frac{\partial t}{\partial x} + y \frac{\partial t}{\partial y} = nt$

Proof: Let $t = x^n \phi\left(\frac{y}{x}\right)$, $\therefore t$ is homogeneous function in x and y of degree n

$$\therefore \frac{\partial t}{\partial x} = x^n \phi'\left(\frac{y}{x}\right) \left(-\frac{y}{x^2}\right) + nx^{n-1} \phi\left(\frac{y}{x}\right)$$

$$\text{and } \frac{\partial t}{\partial y} = x^n \phi' \left(\frac{y}{x} \right) \left(\frac{1}{x} \right)$$

$$\Rightarrow x \frac{\partial t}{\partial x} + y \frac{\partial t}{\partial y} = x^{n-2+1} \phi' \left(\frac{y}{x} \right) (-y) + nx^n \phi \left(\frac{y}{x} \right) + x^{n-1} y \phi' \left(\frac{y}{x} \right) = nx^n \phi \left(\frac{y}{x} \right) = nt$$

Deductions from Euler's Theorem

1. If $t = f(x, y, z)$ be a homogeneous function of degree n ,

$$x \frac{\partial t}{\partial x} + y \frac{\partial t}{\partial y} + z \frac{\partial t}{\partial z} = nt$$

2. If $t = f(x, y)$ be a homogeneous function of degree n ,

$$x^2 \frac{\partial^2 t}{\partial x^2} + 2xy \frac{\partial^2 t}{\partial x \partial y} + y^2 \frac{\partial^2 t}{\partial y^2} = n(n-1)t$$

3. If $t = f(x, y)$ is not a homogenous function, but $u = f(t)$ be a homogenous

function in x and y of degree n , then $x \frac{\partial t}{\partial x} + y \frac{\partial t}{\partial y} = g(t)$, where $g(t) = \frac{nf(t)}{f'(t)}$

4. Again, if $u = f(t)$ be a homogenous function in x and y of degree n ,

$$x^2 \frac{\partial^2 t}{\partial x^2} + 2xy \frac{\partial^2 t}{\partial x \partial y} + y^2 \frac{\partial^2 t}{\partial y^2} = g(t)(g'(t) - 1), \text{ where } g(t) = \frac{nf(t)}{f'(t)}$$

Example3 If $t = x\phi \left(\frac{y}{x} \right) + x^2\psi \left(\frac{y}{x} \right)$, prove that $x^2 \frac{\partial^2 t}{\partial x^2} + 2xy \frac{\partial^2 t}{\partial x \partial y} + y^2 \frac{\partial^2 t}{\partial y^2} = 2$

Solution: Let $t = t_1 + t_2$, where $t_1 = x\phi \left(\frac{y}{x} \right)$, $t_2 = x^2\psi \left(\frac{y}{x} \right)$

$$\text{Now } x^2 \frac{\partial^2 t_1}{\partial x^2} + 2xy \frac{\partial^2 t_1}{\partial x \partial y} + y^2 \frac{\partial^2 t_1}{\partial y^2} = 1(1-1)t_1 = 0$$

$\therefore t_1$ is a homogeneous function of degree one

$$\text{Also } x^2 \frac{\partial^2 t_2}{\partial x^2} + 2xy \frac{\partial^2 t_2}{\partial x \partial y} + y^2 \frac{\partial^2 t_2}{\partial y^2} = 2(2-1)t_2 = 2$$

$\therefore t_2$ is a homogeneous function of degree two

$$\therefore x^2 \frac{\partial^2 (t_1+t_2)}{\partial x^2} + 2xy \frac{\partial^2 (t_1+t_2)}{\partial x \partial y} + y^2 \frac{\partial^2 (t_1+t_2)}{\partial y^2} = 0 + 2 = 2$$

Example4 If $t = \log \frac{x+y}{\sqrt{x^2+y^2}}$, Prove that $x \frac{\partial t}{\partial x} + y \frac{\partial t}{\partial y} = 0$

Solution: Here $t = \log \frac{x+y}{\sqrt{x^2+y^2}}$ is not a homogeneous function of x and y ,

$$u = e^t = \frac{x+y}{\sqrt{x^2+y^2}} = \frac{x(1+\frac{y}{x})}{x\sqrt{1+(\frac{y}{x})^2}} = x^0 \phi\left(\frac{y}{x}\right) \text{ is a homogeneous function of degree } 0$$

$$\therefore x \frac{\partial t}{\partial x} + y \frac{\partial t}{\partial y} = g(t), \text{ where } g(t) = \frac{nf(t)}{f'(t)}$$

$$\Rightarrow x \frac{\partial t}{\partial x} + y \frac{\partial t}{\partial y} = 0 \quad \because n = 0$$

Example5 If $u = \sin^{-1}(x^3 + y^3)^{\frac{2}{3}}$, Prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2$

Solution: Here $u = \sin^{-1}(x^3 + y^3)^{\frac{2}{3}}$ is not a homogeneous function of x and y ,

$$v = \sin u = (x^3 + y^3)^{\frac{2}{3}} = x^2 \left(1 + \frac{y^3}{x^3}\right)^{\frac{2}{3}} \text{ is a homogeneous function of degree } 2$$

$$\therefore x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u)(g'(u) - 1), \text{ where } g(u) = \frac{nf(u)}{f'(u)}$$

$$\text{Here } g(u) = \frac{nf(u)}{f'(u)} = \frac{2 \sin u}{\cos u} = 2 \tan u, \quad g'(u) = 2 \sec^2 u$$

$$\Rightarrow x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \tan u (2 \sec^2 u - 1)$$

Example6 If $t = \tan^{-1} \frac{x^2+y^2}{x-y}$, Prove that $x \frac{\partial t}{\partial x} + y \frac{\partial t}{\partial y} = \frac{1}{2} \sin 2t$

Also prove $x^2 \frac{\partial^2 t}{\partial x^2} + 2xy \frac{\partial^2 t}{\partial x \partial y} + y^2 \frac{\partial^2 t}{\partial y^2} = \frac{1}{2} \sin 2t (\cos 2t - 1)$

Solution: Here $t = \tan^{-1} \frac{x^2+y^2}{x-y}$ is not a homogeneous function of x and y ,

$$u = \tan t = \frac{x^2+y^2}{x-y} = \frac{x^2(1+(\frac{y}{x})^2)}{x(1+\frac{y}{x})} = x \phi\left(\frac{y}{x}\right) \text{ is a homogeneous function of degree } 1$$

$$\therefore x \frac{\partial t}{\partial x} + y \frac{\partial t}{\partial y} = g(t), \text{ where } g(t) = \frac{nf(t)}{f'(t)}$$

$$\text{and } x^2 \frac{\partial^2 t}{\partial x^2} + 2xy \frac{\partial^2 t}{\partial x \partial y} + y^2 \frac{\partial^2 t}{\partial y^2} = g(t)(g'(t) - 1), \text{ where } g(t) = \frac{nf(t)}{f'(t)}$$

$$\text{Here } g(t) = \frac{nf(t)}{f'(t)} = \frac{\tan t}{\sec^2 t} = \sin t \cdot \cos t = \frac{1}{2} \sin 2t, \quad g'(t) = \cos 2t$$

$$\Rightarrow x^2 \frac{\partial^2 t}{\partial x^2} + 2xy \frac{\partial^2 t}{\partial x \partial y} + y^2 \frac{\partial^2 t}{\partial y^2} = \frac{1}{2} \sin 2t (\cos 2t - 1)$$

1.3 Composite Functions

- If $z = f(x, y)$, such that $x = g(t)$ and $y = h(t)$, z is called a composite function of single variable t .
- $z = f(x, y)$, such that $x = g(u, v)$ and $y = h(u, v)$, z is called a composite function of two variables u and v .

1.3.1 Differentiation of Composite Functions (Chain Rule)

- If $z = f(x, y)$, where $x = g(t)$ and $y = h(t) \Rightarrow \frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$
- If $z = f(x, y)$, where $x = g(u, v)$ and $y = h(u, v)$
 $\Rightarrow \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$ and $\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$
- If $z = f(x, y)$, where $y = g(x) \Rightarrow \frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dx}$

Example7 If $z = f(x, y)$, $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$,

$$\text{Prove that } \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$$

Solution: Here $z = f(x, y)$, such that $x = g(u, v)$ and $y = h(u, v)$

$$\begin{aligned} \therefore \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} = \frac{\partial z}{\partial x} e^u - \frac{\partial z}{\partial y} e^{-u} \\ \text{and } \frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v} = -\frac{\partial z}{\partial x} e^{-v} - \frac{\partial z}{\partial y} e^v \\ \Rightarrow \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x} e^u - \frac{\partial z}{\partial y} e^{-u} + \frac{\partial z}{\partial x} e^{-v} + \frac{\partial z}{\partial y} e^v \\ &= (e^u + e^{-v}) \frac{\partial z}{\partial x} - (e^{-u} - e^v) \frac{\partial z}{\partial y} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} \end{aligned}$$

Example8 If $z = f(x, y)$, $x = \log u$ and $y = \log v$,

$$\text{Prove that } \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = u^2 \frac{\partial^2 z}{\partial u^2} + v^2 \frac{\partial^2 z}{\partial v^2} + u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v}$$

Solution: Here $z = f(x, y)$, such that $x = g(u)$ and $y = h(v)$

$$\begin{aligned} \therefore \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} \quad \because \frac{\partial y}{\partial u} = 0 \\ &= \frac{\partial z}{\partial x} \frac{dx}{du} = \frac{\partial z}{\partial x} \cdot \frac{1}{u} \end{aligned}$$

$$\Rightarrow \frac{\partial^2 z}{\partial u^2} = \frac{\partial}{\partial u} \left[\frac{\partial z}{\partial u} \right] = \frac{\partial z}{\partial x} \cdot \left(-\frac{1}{u^2} \right) + \frac{1}{u} \frac{\partial^2 z}{\partial u \partial x}$$

$$\text{Similarly } \frac{\partial z}{\partial v} = \frac{\partial z}{\partial y} \cdot \frac{1}{v} \text{ and } \frac{\partial^2 z}{\partial v^2} = \frac{\partial z}{\partial y} \cdot \left(-\frac{1}{v^2} \right) + \frac{1}{v} \frac{\partial^2 z}{\partial v \partial y}$$

$$\begin{aligned} \therefore u^2 \frac{\partial^2 z}{\partial u^2} + v^2 \frac{\partial^2 z}{\partial v^2} + u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v} &= -\frac{\partial z}{\partial x} + u \frac{\partial^2 z}{\partial u \partial x} - \frac{\partial z}{\partial y} + v \frac{\partial^2 z}{\partial v \partial y} + \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \\ &= u \frac{\partial^2 z}{\partial u \partial x} + v \frac{\partial^2 z}{\partial v \partial y} = u \frac{\partial}{\partial x} \left[\frac{\partial z}{\partial u} \right] + v \frac{\partial}{\partial y} \left[\frac{\partial z}{\partial v} \right] \\ &= u \frac{\partial}{\partial x} \left[\frac{\partial z}{\partial x} \cdot \frac{1}{u} \right] + v \frac{\partial}{\partial y} \left[\frac{\partial z}{\partial y} \cdot \frac{1}{v} \right] = \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} \end{aligned}$$

Example9 If $u = x \log xy$, $x^3 + y^3 + 3xy = 1$, find $\frac{du}{dx}$

Solution: Here $u = f(x, y)$, such that $y = f(x)$

$$\therefore \frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dx} \quad \dots \textcircled{1}$$

$$\text{Now } \frac{\partial u}{\partial x} = x \cdot \frac{y}{xy} + \log xy = 1 + \log xy \quad \dots \textcircled{2}$$

$$\text{Also } \frac{\partial u}{\partial y} = x \cdot \frac{x}{xy} = \frac{x}{y} \quad \dots \textcircled{3}$$

Further to find $\frac{dy}{dx}$, differentiating $x^3 + y^3 + 3xy = 1$ with respect to x

$$\Rightarrow 3x^2 + 3y^2 \frac{dy}{dx} + 3x \frac{dy}{dx} + 3y = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x^2 + y}{y^2 + x} \quad \dots \textcircled{4}$$

Using $\textcircled{2}$, $\textcircled{3}$, $\textcircled{4}$ in $\textcircled{1}$, we get

$$\frac{du}{dx} = 1 + \log xy - \frac{x}{y} \left(\frac{x^2 + y}{y^2 + x} \right)$$

Example10 If $u = f(x^2 + 2yz, y^2 + 2zx)$, prove that

$$(y^2 - zx) \frac{\partial u}{\partial x} + (x^2 - yz) \frac{\partial u}{\partial y} + (z^2 - xy) \frac{\partial u}{\partial z} = 0$$

Solution: Let $p = x^2 + 2yz$, $q = y^2 + 2zx$

$\therefore u = f(p, q)$, such that $p = g(x, y, z)$, $q = h(x, y, z)$

$$\text{Now } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} \cdot \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \cdot \frac{\partial q}{\partial x}$$

$$= \frac{\partial u}{\partial p} \cdot 2x + \frac{\partial u}{\partial q} \cdot 2z \quad \dots \textcircled{1}$$

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial p} \cdot \frac{\partial p}{\partial y} + \frac{\partial u}{\partial q} \cdot \frac{\partial q}{\partial y} \\ &= \frac{\partial u}{\partial p} \cdot 2z + \frac{\partial u}{\partial q} \cdot 2y \quad \dots \textcircled{2} \end{aligned}$$

$$\begin{aligned} \frac{\partial u}{\partial z} &= \frac{\partial u}{\partial p} \cdot \frac{\partial p}{\partial z} + \frac{\partial u}{\partial q} \cdot \frac{\partial q}{\partial z} \\ &= \frac{\partial u}{\partial p} \cdot 2y + \frac{\partial u}{\partial q} \cdot 2x \quad \dots \textcircled{3} \end{aligned}$$

Substituting the values of $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial u}{\partial z}$ from $\textcircled{1}$, $\textcircled{2}$, $\textcircled{3}$ respectively,

$$\text{we get } (y^2 - zx) \frac{\partial u}{\partial x} + (x^2 - yz) \frac{\partial u}{\partial y} + (z^2 - xy) \frac{\partial u}{\partial z} = 0$$

1.4 Differentiation of Implicit Functions

If $f(x, y) = 0$ is an implicit relation between x and y ,

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

$$\Rightarrow \frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{f_x}{f_y}, \text{ also } \frac{d^2y}{dx^2} = -\frac{f_{xx}f_y^2 - 2f_xf_yf_{xy} + f_{yy}f_x^2}{f_y^3}$$

Note: We can evaluate $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for implicit functions by direct differentiation as well

Example10 If $x^3 + 3x^2y + 6xy^2 + y^3 = 1$, Find $\frac{dy}{dx}$ using differential formula of implicit functions. Also verify the result by direct differentiation.

Solution: For implicit functions, $\frac{dy}{dx} = -\frac{f_x}{f_y}$

$$\text{Here } f_x = 3x^2 + 6xy + 6y^2, f_y = 3x^2 + 12xy + 3y^2$$

$$\therefore \frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{x^2 + 2xy + 2y^2}{x^2 + 4xy + y^2}$$

Verification: Differentiating the equation $x^3 + 3x^2y + 6xy^2 + y^3 = 1$ w.r.t. x

$$\Rightarrow 3x^2 + 3x^2 \frac{dy}{dx} + 6xy + 12xy \frac{dy}{dx} + 6y^2 + 3y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x^2+2xy+2y^2}{x^2+4xy+y^2}$$

Example11 If $x^n + y^n = a^n$, Find $\frac{d^2y}{dx^2}$

Solution: For implicit functions $\frac{d^2y}{dx^2} = -\frac{f_{xx}f_y^2 - 2f_xf_yf_{xy} + f_{yy}f_x^2}{f_y^3}$

$$\text{Here } f_x = nx^{n-1}, f_{xx} = n(n-1)x^{n-2}, f_x^2 = n^2x^{2n-2}, f_{xy} = 0$$

$$f_y = ny^{n-1}, f_{yy} = n(n-1)y^{n-2}, f_y^2 = n^2y^{2n-2}, f_y^3 = n^3y^{3n-3}$$

$$\begin{aligned} \therefore \frac{d^2y}{dx^2} &= -\frac{n(n-1)x^{n-2} \cdot n^2y^{2n-2} - 2nx^{n-1}ny^{n-1} \cdot 0 + n(n-1)y^{n-2} \cdot n^2x^{2n-2}}{n^3y^{3n-3}} \\ &= -\frac{n^3(n-1)x^{n-2}y^{2n-2} + n^3(n-1)y^{n-2}x^{2n-2}}{n^3y^{3n-3}} \\ &= -n^3(n-1)x^{n-2}y^{n-2} \frac{y^n + x^n}{n^3y^{3n-3}} \\ &= -\frac{(n-1)x^{n-2}a^n}{y^{2n-1}} \end{aligned}$$

1.5 Exact Differentials

A differential of the form $df = f_1(x, y)dx + f_2(x, y)dy$ is said to be exact or perfect,

$$\text{if } \frac{\partial f_1(x, y)}{\partial y} = \frac{\partial f_2(x, y)}{\partial x}$$

Example12 Given the differential $df = (x^2 + 4xy + 3y^2)dx + (2x^2 + 6xy)dy$, check whether it is exact or not.

Solution: Given differential is of the form $df = f_1(x, y)dx + f_2(x, y)dy$,

$$f_1(x, y) = x^2 + 4xy + 3y^2, f_2(x, y) = 2x^2 + 6xy$$

$$\text{We have } \frac{\partial f_1(x, y)}{\partial y} = \frac{\partial(x^2+4xy+3y^2)}{\partial y} = 4x + 6y \quad \dots \textcircled{1}$$

$$\text{Also } \frac{\partial f_2(x, y)}{\partial x} = \frac{\partial(2x^2+6xy)}{\partial x} = 4x + 6y \quad \dots \textcircled{2}$$

$$\text{From } \textcircled{1} \text{ and } \textcircled{2} \quad \frac{\partial f_1(x, y)}{\partial y} = \frac{\partial f_2(x, y)}{\partial x}$$

$\therefore df = (x^2 + 4xy + 3y^2)dx + (2x^2 + 6xy)dy$ is an exact differential.

1.6 Maxima and Minima of Functions of Two Variables

A function $z = f(x, y)$ has extreme points, where both the partial derivatives $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ are zero. An extreme point of a function of two variables variable is either a local maximum, a local minimum, or a saddle point.

Saddle Point: A point (x_i, y_i) on the curve $z = f(x, y)$, where $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ are zero, but the value of function $z = f(x, y)$ on L.H.S. and R.H.S. of point (x_i, y_i) are of different signs. Here the function does not have any maximum or minimum value, but a point of inflection.

1.6.1 Algorithm to find maximum or minimum values of a function $z = f(x, y)$

1. Find the partial derivatives $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
2. Solve the equations $\frac{\partial z}{\partial x} = 0$ and $\frac{\partial z}{\partial y} = 0$, simultaneously to obtain the extreme points as (x_i, y_i) .
3. Find the differentials $\frac{\partial^2 z}{\partial x^2} \equiv r$, $\frac{\partial^2 z}{\partial x \partial y} \equiv s$ and $\frac{\partial^2 z}{\partial y^2} \equiv t$
4. Evaluate the expression $(rt - s^2)$ at extreme points (x_i, y_i) .
 - i. If $[rt - s^2]_{(x_i, y_i)} > 0$, check the value of $[r]_{(x_i, y_i)}$
 - a. $[r]_{(x_i, y_i)} > 0 \Rightarrow z = f(x, y)$ has a minimum value at (x_i, y_i)
 - b. $[r]_{(x_i, y_i)} < 0 \Rightarrow z = f(x, y)$ has a maximum value at (x_i, y_i)
 - ii. If $[rt - s^2]_{(x_i, y_i)} = 0$, check the value of $z = f(x_i, y_i)$ in the immediate vicinity on both sides. If the value of function $z = f(x, y)$ on L.H.S. and R.H.S. of point (x_i, y_i) are of different signs, it is a saddle point.
 - iii. If $[rt - s^2]_{(x_i, y_i)} < 0 \Rightarrow z = f(x, y)$ has no extreme value at (x_i, y_i) , i.e., it is a saddle point.

Example13 Locate the stationary points of $u = x^4 + y^4 - 2x^2 + 4xy - 2y^2$ and determine their nature.

Solution: For stationary points $\frac{\partial u}{\partial x} = 0$ and $\frac{\partial u}{\partial y} = 0$

$$\Rightarrow \frac{\partial u}{\partial x} = 4x^3 - 4x + 4y = 0 \quad \dots \textcircled{1}$$

$$\frac{\partial u}{\partial y} = 4y^3 + 4x - 4y = 0 \quad \dots \textcircled{2}$$

Adding $\textcircled{1}$ and $\textcircled{2}$, we get

$$x^3 + y^3 = 0$$

$$\Rightarrow (x + y)(x^2 - xy + y^2) = 0$$

$$\Rightarrow (x + y) = 0 \quad \text{or} \quad (x^2 - xy + y^2) = 0$$

$$\Rightarrow y = -x \quad \dots \textcircled{3} \quad \left| \begin{array}{l} \Rightarrow x^2 - 2xy + y^2 = -xy \\ \Rightarrow (x - y)^2 = -xy \\ \text{which is not possible} \end{array} \right.$$

Using $\textcircled{3}$ in $\textcircled{1}$, we get $4x^3 - 4x - 4x = 0$

$$\Rightarrow 4x^3 - 8x = 0 \Rightarrow x(x^2 - 2) = 0$$

$$\Rightarrow x = 0, \pm\sqrt{2}$$

$\therefore$ The extreme points using $\textcircled{3}$ are $(0,0)$, $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$

$$\text{Now } \frac{\partial^2 u}{\partial x^2} \equiv r = 12x^2 - 4, \quad \frac{\partial^2 u}{\partial x \partial y} \equiv s = 4, \quad \frac{\partial^2 u}{\partial y^2} \equiv t = 12y^2 - 4$$

$$\therefore rt - s^2 = 4(3x^2 - 1)4(3y^2 - 1) - 16 = 16[(3x^2 - 1)(3y^2 - 1) - 1]$$

Again, $[rt - s^2]_{(0,0)} = 0$, checking the values of the function $u = f(x, y)$ in the immediate vicinity on both sides.

$$[u]_{(-1,1)} = 1 + 1 - 2 - 4 - 2 < 0, \quad [u]_{(1,1)} = 1 + 1 - 2 + 4 - 2 > 0$$

$\therefore (0,0)$ is a saddle point

$$\text{Again } [rt - s^2]_{(\sqrt{2}, -\sqrt{2})} = 16(5)(5) - 16 > 0, \text{ and } [r]_{(\sqrt{2}, -\sqrt{2})} > 0$$

$\therefore (\sqrt{2}, -\sqrt{2})$ is a point of minima

$$\text{Also } [rt - s^2]_{(-\sqrt{2}, \sqrt{2})} > 0, \text{ and } [r]_{(-\sqrt{2}, \sqrt{2})} > 0$$

$\therefore (-\sqrt{2}, \sqrt{2})$ is also a point of minima

Example14 Locate the stationary points of the function $\sin x + \sin y + \sin(x + y)$, where $x, y \in [0, \pi]$.

Solution: Let $f(x, y) = \sin x + \sin y + \sin(x + y)$

For stationary points $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$

$$\Rightarrow \frac{\partial f}{\partial x} = \cos x + \cos(x + y) = 0 \quad \dots \textcircled{1}$$

$$\frac{\partial f}{\partial y} = \cos y + \cos(x + y) = 0 \quad \dots \textcircled{2}$$

Subtracting $\textcircled{2}$ from $\textcircled{1}$, we get

$$\cos x - \cos y = 0$$

$$\Rightarrow \cos x = \cos y$$

$$\Rightarrow x = y \quad \because x, y \in [0, \pi] \quad \dots \textcircled{3}$$

Using $\textcircled{3}$ in $\textcircled{1}$, we get

$$\cos x + \cos 2x = 0$$

$$\Rightarrow \cos 2x = -\cos x$$

$$\Rightarrow \cos 2x = \cos(\pi - x)$$

$$\Rightarrow 2x = 2n\pi \pm (\pi - x), n \in \mathbb{Z}$$

$$\Rightarrow 3x = (2n + 1)\pi \text{ or } x = (2n - 1)\pi, n \in \mathbb{Z}$$

$$\Rightarrow x = \frac{\pi}{3}, \pi \text{ for } n = 0 \text{ and } n = 1 \text{ respectively} \quad \because x \in [0, \pi]$$

$\therefore$ The extreme points using $\textcircled{3}$ are $(\frac{\pi}{3}, \frac{\pi}{3})$ and (π, π) respectively.

$$\text{Now } \frac{\partial^2 f}{\partial x^2} \equiv r = -\sin x - \sin(x + y)$$

$$\frac{\partial^2 f}{\partial x \partial y} \equiv s = -\sin(x + y)$$

$$\frac{\partial^2 f}{\partial y^2} \equiv t = -\sin y - \sin(x + y)$$

$$\text{Now } [r]_{\left(\frac{\pi}{3}, \frac{\pi}{3}\right)} = -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = -\sqrt{3} < 0$$

$$[s]_{\left(\frac{\pi}{3}, \frac{\pi}{3}\right)} = -\sin\left(\frac{\pi}{3} + \frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$[t]_{\left(\frac{\pi}{3}, \frac{\pi}{3}\right)} = -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = -\sqrt{3}$$

$$\therefore [rt - s^2]_{\left(\frac{\pi}{3}, \frac{\pi}{3}\right)} = (-\sqrt{3})(-\sqrt{3}) - \left(-\frac{\sqrt{3}}{2}\right)^2 = 3 - \left(-\frac{\sqrt{3}}{2}\right)^2 > 0$$

$$\Rightarrow [rt - s^2]_{\left(\frac{\pi}{3}, \frac{\pi}{3}\right)} > 0 \text{ and } [r]_{\left(\frac{\pi}{3}, \frac{\pi}{3}\right)} < 0$$

$$\therefore f(x, y) \text{ has a maximum value at } \left(\frac{\pi}{3}, \frac{\pi}{3}\right)$$

$$\text{Also } f\left(\frac{\pi}{3}, \frac{\pi}{3}\right) = \sin\frac{\pi}{3} + \sin\frac{\pi}{3} + \sin\left(\frac{\pi}{3} + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

$$\therefore \text{Maximum value of the function is } \frac{3\sqrt{3}}{2} \text{ at the point } \left(\frac{\pi}{3}, \frac{\pi}{3}\right)$$

$$\text{Also } [r]_{(\pi, \pi)} = 0$$

$$[s]_{(\pi, \pi)} = 0$$

$$[t]_{(\pi, \pi)} = 0$$

$$\therefore [rt - s^2]_{(\pi, \pi)} = 0$$

Checking the values of the function $f(x, y)$ in the immediate vicinity on both sides.

$$[f]_{\left(\frac{5\pi}{6}, \frac{5\pi}{6}\right)} = \sin\frac{5\pi}{6} + \sin\frac{5\pi}{6} + \sin\frac{10\pi}{6} = \frac{1}{2} + \frac{1}{2} - \frac{\sqrt{3}}{2} > 0$$

$$\text{Also } [f]_{\left(\frac{7\pi}{6}, \frac{7\pi}{6}\right)} = \sin\frac{7\pi}{6} + \sin\frac{7\pi}{6} + \sin\frac{14\pi}{6} = -\frac{1}{2} - \frac{1}{2} + \frac{\sqrt{3}}{2} < 0$$

$$\therefore (\pi, \pi) \text{ is a saddle point}$$

1.7 Method of Lagrange Multipliers for Finding Extreme Values

If we want to find extreme values of a function whose variables are connected by a relation, we can use the method of Lagrange Multipliers.

1.7.1 Algorithm to find maximum or minimum values of a function $f(x, y, z)$ using Lagrange Multipliers

1. Consider the function: Maximize/Minimize $f(x, y, z)$ subject to $g(x, y, z) = k$
2. Form an auxiliary function $F(x, y, z, \lambda) = f(x, y, z) + \lambda[g(x, y, z) - k]$
3. Solve the equations $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0$ and $g(x, y, z) = k$ by the method of grouping or method of multipliers or both.

***Method of multipliers:** Consider a fraction $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$
 Taking x, y, z as multipliers, each fraction $= \frac{1 \times x}{2 \times x} = \frac{2 \times y}{4 \times y} = \frac{3 \times z}{6 \times z}$

Example 15 Find the extreme values of $f(x, y) = 3x^2 + 4y^2$, given that $x + y = 35$

Solution: Let $F(x, y, \lambda) = 3x^2 + 4y^2 + \lambda(x + y - 35)$

$$\Rightarrow F(x, y, \lambda) = 3x^2 + 4y^2 + \lambda x + \lambda y - 35\lambda$$

Differentiating $F(x, y, \lambda)$ with respect to x & y , and equating to zero

$$\frac{\partial F}{\partial x} = 6x + \lambda x = 0 \Rightarrow x = -\frac{\lambda}{6} \quad \dots \textcircled{1}$$

$$\frac{\partial F}{\partial y} = 8y + \lambda y = 0 \Rightarrow y = -\frac{\lambda}{8} \quad \dots \textcircled{2}$$

$$\text{Given } x + y = 35 \quad \dots \textcircled{3}$$

Using $\textcircled{1}$ and $\textcircled{2}$ in $\textcircled{3}$, we get

$$-\frac{\lambda}{6} - \frac{\lambda}{8} = 35 \Rightarrow -\frac{7\lambda}{24} = 35 \quad \therefore \lambda = -120$$

Substituting the value of λ in $\textcircled{1}$ and $\textcircled{2}$

$$x = -\frac{\lambda}{6} = \frac{120}{6} = 20, \quad y = -\frac{\lambda}{8} = \frac{120}{8} = 15$$

$\therefore$ The extreme value of the function is $f(20, 15) = 3(20)^2 + 4(15)^2 = 2100$

Checking the value at nearby point $f(19, 16) = 3(19)^2 + 4(16)^2 = 2107$

$\Rightarrow (20, 15)$ is the minima of the function $f(x, y) = 3x^2 + 4y^2$, subject to $x + y = 35$

Example 16 Find the extreme values of $f(x, y, z) = xyz$, subject to $x^2 + y^2 + z^2 = 3$

Solution: Let $F(x, y, z, \lambda) = xyz + \lambda(x^2 + y^2 + z^2 - 3)$

$$\Rightarrow F(x, y, z, \lambda) = xyz + \lambda x^2 + \lambda y^2 + \lambda z^2 - 3\lambda$$

Differentiating $F(x, y, z, \lambda)$ with respect to x, y, z and equating to zero

$$\frac{\partial F}{\partial x} = yz + 2\lambda x = 0 \Rightarrow \lambda = -\frac{yz}{2x} \quad \dots \textcircled{1}$$

$$\frac{\partial F}{\partial y} = xz + 2\lambda y = 0 \Rightarrow \lambda = -\frac{xz}{2y} \quad \dots \textcircled{2}$$

$$\frac{\partial F}{\partial z} = xy + 2\lambda z = 0 \Rightarrow \lambda = -\frac{xy}{2z} \quad \dots \textcircled{3}$$

From $\textcircled{1}$, $\textcircled{2}$ and $\textcircled{3}$, we get

$$\frac{yz}{2x} = \frac{xz}{2y} = \frac{xy}{2z}$$

$$\text{Taking } x, y, z \text{ as multipliers} \Rightarrow \frac{xyz}{2x^2} = \frac{xyz}{2y^2} = \frac{xyz}{2z^2} \Rightarrow x^2 = y^2 = z^2 \quad \dots \textcircled{4}$$

$$\text{Given } x^2 + y^2 + z^2 = 3 \quad \dots \textcircled{5}$$

Using $\textcircled{4}$ in $\textcircled{5}$, we get

$$3x^2 = 3 \Rightarrow x = \pm 1, y = \pm 1, z = \pm 1$$

$\therefore (1,1,1), (1,-1,1), (1,1,-1), (1,-1,-1), (-1,1,1), (-1,-1,1), (-1,1,-1), (-1,-1,-1)$ are the eight points of maxima or minima.

Clearly $f(x, y, z) = 1$ at $(1,1,1), (1,-1,-1), (-1,-1,1)$ and $(-1,1,-1)$

$\therefore (1,1,1), (1,-1,-1), (-1,-1,1)$ and $(-1,1,-1)$ are the points of maxima

Also $f(x, y, z) = -1$ at $(1,-1,1), (1,1,-1), (-1,1,1)$ and $(-1,-1,-1)$

$\therefore (1,-1,1), (1,1,-1), (-1,1,1)$ and $(-1,-1,-1)$ are the points of minima

Example17 Find the dimensions of the box with maximum volume if the total surface area is 24cm^2 .

Solution: Let $V(x, y, z) = xyz$, given that $2xy + 2yz + 2zx = 24$

$$\text{Let } F(x, y, z, \lambda) = xyz + \lambda(xy + yz + zx - 12)$$

$$\Rightarrow F(x, y, z, \lambda) = xyz + \lambda xy + \lambda yz + \lambda zx - 12\lambda$$

Differentiating $F(x, y, z, \lambda)$ with respect to x, y, z and equating to zero

$$\frac{\partial F}{\partial x} = yz + \lambda y + \lambda z = 0 \Rightarrow \lambda = -\frac{yz}{y+z} \quad \dots \textcircled{1}$$

$$\frac{\partial F}{\partial y} = zx + \lambda x + \lambda z = 0 \Rightarrow \lambda = -\frac{xz}{x+z} \quad \dots \textcircled{2}$$

$$\frac{\partial F}{\partial z} = xy + \lambda y + \lambda x = 0 \Rightarrow \lambda = -\frac{xy}{x+y} \quad \dots \textcircled{3}$$

From $\textcircled{1}$, $\textcircled{2}$ and $\textcircled{3}$, we get

$$\frac{yz}{y+z} = \frac{xz}{x+z} = \frac{xy}{x+y}$$

$$\text{Taking } x, y, z \text{ as multipliers} \Rightarrow \frac{xyz}{x(y+z)} = \frac{xyz}{y(x+z)} = \frac{xyz}{z(x+y)}$$

From first two equations, we get

$$\Rightarrow x(y+z) = y(x+z)$$

$$\Rightarrow xy + xz = xy + yz \Rightarrow x = y \quad \dots \textcircled{4}$$

Similarly from last two equations, we get $y = z \quad \dots \textcircled{5}$

From $\textcircled{4}$ and $\textcircled{5}$, we get $x = y = z \quad \dots \textcircled{6}$

$$\text{Given } xy + yz + zx = 12 \quad \dots \textcircled{7}$$

Using $\textcircled{6}$ in $\textcircled{7}$, we get

$$3x^2 = 12 \Rightarrow x^2 = 4$$

$$\therefore x = 2, y = 2, z = 2 \quad \because x = y = z \text{ and } x, y, z > 0$$

Hence maximum volume of the cube is $V(x, y, z) = xyz = 8$ cubic units

1.8 Differentiation Under Integral Sign

Differentiation under the integral sign is a rule which is used to compute some specific integrals with more than one variable. The variable other than variable of integration is taken as a parameter, and the integral is differentiated with respect to the parameter.

Leibnitz's Rule for Differentiation Under Integral Sign

$$\text{If } I = \int_{a(t)}^{b(t)} f(x, t) dx, \text{ then } \frac{dI}{dt} = \int_{a(t)}^{b(t)} \frac{\partial f(x, t)}{\partial t} dx + f(b(t), t) \frac{db}{dt} - f(a(t), t) \frac{da}{dt}$$

Deduction: If the limits a and b are constants (independent of t), $\frac{dI(t)}{dt} = \int_a^b \frac{\partial f(x, t)}{\partial t} dx$

Note: Here t is the parameter because integration is being performed with respect to x .

Example 18 If $I = \int_0^{a^2} \tan^{-1} \frac{x}{a} dx$, prove that $\frac{dI}{da} = -\frac{1}{2} \log(1 + a^2) + 2a \tan^{-1} a$

Solution: Given $I = \int_0^{a^2} \tan^{-1} \frac{x}{a} dx$... ①

Differentiating ① with respect to parameter a

$$\begin{aligned} \frac{dI}{da} &= \int_0^{a^2} \frac{\partial}{\partial a} \left(\tan^{-1} \frac{x}{a} \right) dx + f(a^2, a) \frac{da^2}{da} - f(0, a) \frac{d(0)}{da} \quad \text{by Leibnitz's Rule} \\ \Rightarrow \frac{dI}{da} &= \int_0^{a^2} \frac{1}{1 + \frac{x^2}{a^2}} \left(-\frac{x}{a^2} \right) dx + \tan^{-1} \frac{a^2}{a} (2a) - 0 \\ &= \int_0^{a^2} -\frac{x}{a^2 + x^2} dx + 2a \tan^{-1} a \\ &= -\frac{1}{2} \int_0^{a^2} \frac{2x}{a^2 + x^2} dx + 2a \tan^{-1} a \\ &= -\frac{1}{2} [\log(a^2 + x^2)]_0^{a^2} + 2a \tan^{-1} a \quad \because \int \frac{f'(t)}{f(t)} dx = \log f(t) \\ &= -\frac{1}{2} [\log(a^2 + a^4) - \log a^2] + 2a \tan^{-1} a \\ &= -\frac{1}{2} \log \frac{a^2 + a^4}{a^2} + 2a \tan^{-1} a \quad \because \log m - \log n = \log \frac{m}{n} \\ \Rightarrow \frac{dI}{da} &= -\frac{1}{2} \log(1 + a^2) + 2a \tan^{-1} a \end{aligned}$$

Hence proved

Example 19 If $I = \int_{x^2}^{x^3} \frac{1}{\log t} dt$, prove that $\frac{dI}{dx} = \frac{x(x-1)}{\log x}$

Solution: Given $I = \int_{x^2}^{x^3} \frac{1}{\log t} dt$... ①

Differentiating ① with respect to parameter x

$$\begin{aligned} \frac{dI}{dx} &= \int_{x^2}^{x^3} \frac{\partial}{\partial x} \left(\frac{1}{\log t} \right) dt + f(x^3, x) \frac{dx^3}{dx} - f(x^2, x) \frac{dx^2}{dx} \quad \text{by Leibnitz's Rule} \\ \Rightarrow \frac{dI}{dx} &= \int_{x^2}^{x^3} 0 dt + \frac{1}{\log x^3} (3x^2) - \frac{1}{\log x^2} (2x) \\ &= 0 + \frac{3x^2}{3 \log x} - \frac{2x}{2 \log x} = \frac{x^2}{\log x} - \frac{x}{\log x} \quad \because \log m^n = n \log m \\ \Rightarrow \frac{dI}{dx} &= \frac{x(x-1)}{\log x} \quad \text{Hence proved} \end{aligned}$$

Example 20 Prove that $\int_0^\infty \frac{e^{-ax} \sin \lambda x}{x} dx = \tan^{-1} \frac{\lambda}{a}$, hence deduce that $\int_0^\infty \frac{\sin \lambda x}{x} dx = \frac{\pi}{2}$

Solution: Let $I = \int_0^\infty \frac{e^{-ax} \sin \lambda x}{x} dx \quad \dots \textcircled{1}$

Differentiating $\textcircled{1}$ with respect to parameter λ

$$\begin{aligned} \frac{dI}{d\lambda} &= \int_0^\infty \frac{\partial}{\partial \lambda} \left(\frac{e^{-ax} \sin \lambda x}{x} \right) dx \quad \because \text{limits } a \text{ and } b \text{ are independent of parameter } \lambda \\ \Rightarrow \frac{dI}{d\lambda} &= \int_0^\infty \frac{e^{-ax}}{x} \frac{\partial}{\partial \lambda} (\sin \lambda x) dx = \int_0^\infty \frac{e^{-ax}}{x} (x \cos \lambda x) dx \\ &= \int_0^\infty e^{-ax} \cos \lambda x dx = \left[\frac{e^{-ax}}{a^2 + \lambda^2} (-a \cos \lambda x + \lambda \sin \lambda x) \right]_0^\infty \\ &\quad \because \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) \end{aligned}$$

$$\Rightarrow \frac{dI}{d\lambda} = \frac{a}{a^2 + \lambda^2} \quad \because e^{-\infty} = 0$$

$$\Rightarrow dI = \frac{a}{a^2 + \lambda^2} d\lambda$$

Integrating both sides with respect to λ

$$\Rightarrow I = a \cdot \frac{1}{a} \tan^{-1} \frac{\lambda}{a} + c = \tan^{-1} \frac{\lambda}{a} + c$$

If $\lambda = 0 \Rightarrow I = 0$, $\therefore c = 0$ using $\textcircled{1}$

$$\Rightarrow I = \tan^{-1} \frac{\lambda}{a}$$

$$\Rightarrow \int_0^\infty \frac{e^{-ax} \sin \lambda x}{x} dx = \tan^{-1} \frac{\lambda}{a} \quad \dots \textcircled{2}$$

Hence Proved.

Putting $a = 0$ in $\textcircled{2}$, we get

$$\int_0^\infty \frac{\sin \lambda x}{x} dx = \frac{\pi}{2} \quad \because \tan^{-1} \infty = \frac{\pi}{2}$$

Example 21 Prove that $\int_0^{\frac{\pi}{2}} \frac{\log(1+y \sin^2 x)}{\sin^2 x} dx = \pi(\sqrt{1+y} - 1)$

Solution: Let $I = \int_0^{\frac{\pi}{2}} \frac{\log(1+y \sin^2 x)}{\sin^2 x} dx \quad \dots \textcircled{1}$

Differentiating $\textcircled{1}$ with respect to parameter y

$$\begin{aligned}
\frac{dI}{dy} &= \int_0^{\frac{\pi}{2}} \frac{\partial}{\partial y} \left(\frac{\log(1+y \sin^2 x)}{\sin^2 x} \right) dx \quad \because \text{limits } a \text{ and } b \text{ are independent of parameter } y \\
\Rightarrow \frac{dI}{dy} &= \int_0^{\frac{\pi}{2}} \frac{1}{\sin^2 x} \frac{\partial}{\partial \lambda} (\log(1+y \sin^2 x)) dx \\
&= \int_0^{\frac{\pi}{2}} \frac{1}{\sin^2 x} \cdot \frac{\sin^2 x}{(1+y \sin^2 x)} dx = \int_0^{\frac{\pi}{2}} \frac{1}{(1+y \sin^2 x)} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{1}{(\sin^2 x + \cos^2 x + y \sin^2 x)} dx = \int_0^{\frac{\pi}{2}} \frac{1}{\cos^2 x + \sin^2 x(1+y)} dx = \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{1+\tan^2 x(1+y)} dx \\
\Rightarrow \frac{dI}{dy} &= \int_0^{\infty} \frac{1}{1+t^2(1+y)} dt, \text{ by putting } \tan^2 x = t, \sec^2 x dx = dt \\
&= \frac{1}{1+y} \int_0^{\infty} \frac{1}{\frac{1}{(1+y)} + t^2} dt = \frac{1}{1+y} \int_0^{\infty} \frac{1}{\left(\frac{1}{\sqrt{1+y}}\right)^2 + t^2} dt = \frac{\sqrt{1+y}}{1+y} \left[\tan^{-1} \frac{t}{\frac{1}{\sqrt{1+y}}} \right]_0^{\infty} \\
\Rightarrow \frac{dI}{dy} &= \frac{1}{\sqrt{1+y}} (\tan^{-1} \infty - \tan^{-1} 0) = \frac{\pi}{2} \frac{1}{\sqrt{1+y}} \quad \because \tan^{-1} \infty = \frac{\pi}{2} \\
\Rightarrow dI &= \frac{\pi}{2} \frac{1}{\sqrt{1+y}} dy
\end{aligned}$$

Integrating both sides with respect to y

$$\Rightarrow I = \frac{\pi}{2} \cdot 2\sqrt{1+y} + c = \pi\sqrt{1+y} + c \quad \dots \textcircled{2}$$

If $y = 0 \Rightarrow I = 0$, by using $\textcircled{1}$

$\therefore 0 = \pi + c \Rightarrow c = -\pi$, by using $\textcircled{2}$

$$\Rightarrow I = \pi\sqrt{1+y} - \pi = \pi(\sqrt{1+y} - 1)$$

Hence Proved.

1.9 Jacobians and Transformations of Coordinates

We have been using Cartesian coordinate system in most of the applications, where the position of each data point is defined by (x, y) coordinates in two dimensions or by (x, y, z) coordinates in three dimensions. However, for some applications involving curved lines or surfaces, system derived from circular shapes, such as polar, cylindrical or spherical coordinate systems are better used to specify the positions.

1.9.1 Polar Coordinates: A polar coordinate system is a two dimensional system where every point in the plane is determined by its distance r from the origin and an angle θ . Here θ is the angle by which we rotate about the positive direction of x -axis to get to the point. We can establish a relation between Cartesian and polar coordinates using Figure1.

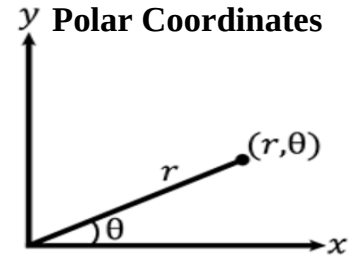


Figure 1

Clearly $x = r \cos \theta$, $y = r \sin \theta$

$$\Rightarrow r = \sqrt{x^2 + y^2} \text{ and } \theta = \tan^{-1} \frac{y}{x}$$

We can convert from Cartesian coordinates to polar and vice-versa using above relations.

1.9.2 Cylindrical Coordinates: In cylindrical coordinate system, xy plane is replaced by the polar plane and the vertical z -axis remains the same (Figure 2).

Here, $x = r \cos \theta$, $y = r \sin \theta$, $z = z$

$$\Rightarrow r = \sqrt{x^2 + y^2} , \theta = \tan^{-1} \frac{y}{x} , z = z$$

We can convert from Cartesian coordinates to cylindrical and vice- versa using above relations.

Cylindrical Coordinates

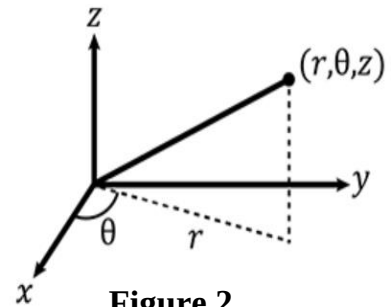


Figure 2

1.9.3 Spherical Polar Coordinates: In spherical coordinate system, we define the position of a point by three coordinates ρ , θ and ϕ (Figure 3). Here ρ ($\rho > 0$) is the distance of the point from the origin (similar to r in polar coordinates), θ is same as that of polar or cylindrical coordinates, and ϕ ($0 \leq \phi \leq \pi$) is the angle that we need to rotate down from the positive z -axis to get to the point.

Spherical Coordinates

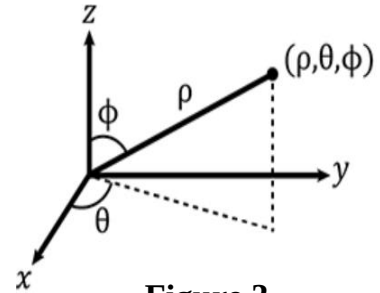


Figure 3

Here $r = \rho \sin \phi$, $z = \rho \cos \phi$, $\rho = \sqrt{r^2 + z^2}$... ①

Also, $x = r \cos \theta$, $y = r \sin \theta$, $z = z$ from cylindrical coordinates ...②

Using ① in ②, we get $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$

$$\Rightarrow \rho = \sqrt{x^2 + y^2 + z^2} , \theta = \tan^{-1} \frac{y}{x} , \phi = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}$$

We can convert from Cartesian coordinates to spherical and vice-versa using above relations.

Note: Some authors define spherical polar coordinates in terms of r , θ and ϕ

Then $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$

Example22 Convert the point $(\sqrt{3}, \frac{\pi}{4}, 1)$ from cylindrical to spherical coordinates.

Solution: We have $\rho^2 = r^2 + z^2 \Rightarrow \rho^2 = 3 + 1 = 4, \therefore \rho = 2$

Also, $\theta = \frac{\pi}{4} \quad \because \theta$ is same in polar, cylindrical and spherical coordinates

Again, $z = \rho \cos \phi \Rightarrow 1 = 2 \cos \phi \Rightarrow \cos \phi = \frac{1}{2} \Rightarrow \phi = \cos^{-1} \frac{1}{2} = \frac{\pi}{3}$

$\therefore$ The point $(\sqrt{3}, \frac{\pi}{4}, 1)$ in cylindrical coordinates is same as the point the point $(2, \frac{\pi}{4}, \frac{\pi}{3})$ in spherical coordinates.

Example23 If $z = f(x, y)$, by changing the independent variables u and v to x and y by means of relations $x = u \cos \alpha - v \sin \alpha, y = u \sin \alpha + v \cos \alpha$,

show that $\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}$ is transformed to $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}$.

Solution: Here $z = f(x, y)$, let $x = g(u, v)$ and $y = h(u, v)$

$$\text{Now } \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} = \cos \alpha \frac{\partial z}{\partial x} + \sin \alpha \frac{\partial z}{\partial y}$$

$$\Rightarrow \frac{\partial}{\partial u}(z) = \left(\cos \alpha \frac{\partial}{\partial x} + \sin \alpha \frac{\partial}{\partial y} \right)(z)$$

$$\Rightarrow \frac{\partial}{\partial u} = \left(\cos \alpha \frac{\partial}{\partial x} + \sin \alpha \frac{\partial}{\partial y} \right)$$

$$\text{Again } \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v} = -\sin \alpha \frac{\partial z}{\partial x} + \cos \alpha \frac{\partial z}{\partial y}$$

$$\Rightarrow \frac{\partial}{\partial v}(z) = \left(-\sin \alpha \frac{\partial}{\partial x} + \cos \alpha \frac{\partial}{\partial y} \right)(z)$$

$$\Rightarrow \frac{\partial}{\partial v} = \left(-\sin \alpha \frac{\partial}{\partial x} + \cos \alpha \frac{\partial}{\partial y} \right)$$

$$\begin{aligned} \therefore \frac{\partial^2 z}{\partial u^2} &= \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} \right) = \left(\cos \alpha \frac{\partial}{\partial x} + \sin \alpha \frac{\partial}{\partial y} \right) \left(\cos \alpha \frac{\partial z}{\partial x} + \sin \alpha \frac{\partial z}{\partial y} \right) \\ &= \cos^2 \alpha \frac{\partial^2 z}{\partial x^2} + 2 \sin \alpha \cos \alpha \frac{\partial^2 z}{\partial x \partial y} + \sin^2 \alpha \frac{\partial^2 z}{\partial y^2} \quad \dots \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{Also, } \frac{\partial^2 z}{\partial v^2} &= \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial v} \right) = \left(-\sin \alpha \frac{\partial}{\partial x} + \cos \alpha \frac{\partial}{\partial y} \right) \left(-\sin \alpha \frac{\partial z}{\partial x} + \cos \alpha \frac{\partial z}{\partial y} \right) \\ &= \sin^2 \alpha \frac{\partial^2 z}{\partial x^2} - 2 \sin \alpha \cos \alpha \frac{\partial^2 z}{\partial x \partial y} + \cos^2 \alpha \frac{\partial^2 z}{\partial y^2} \quad \dots \textcircled{2} \end{aligned}$$

Adding $\textcircled{1}$ and $\textcircled{2}$, we get $\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} = \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}$

Example24 Transform $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ into polar coordinates.

Solution: Let $x = r \cos \theta, y = r \sin \theta \Rightarrow r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \frac{y}{x}$

$$\begin{aligned}
\text{Now } \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial x} = \frac{1}{2} \cdot \frac{2x}{\sqrt{x^2+y^2}} \frac{\partial u}{\partial r} + \frac{1}{\left(1+\left(\frac{y}{x}\right)^2\right)} \left(-\frac{y}{x^2}\right) \frac{\partial u}{\partial \theta} \\
&= \frac{x}{\sqrt{x^2+y^2}} \frac{\partial u}{\partial r} - \frac{y}{x^2+y^2} \frac{\partial u}{\partial \theta} \\
&= \frac{x}{r} \frac{\partial u}{\partial r} - \frac{y}{r^2} \frac{\partial u}{\partial \theta} \quad \because r = \sqrt{x^2+y^2} \\
&= \cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \quad \because x = r \cos \theta \text{ and } y = r \sin \theta
\end{aligned}$$

$$\Rightarrow \frac{\partial}{\partial x} (u) = \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right) (u)$$

$$\Rightarrow \frac{\partial}{\partial x} = \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right)$$

$$\begin{aligned}
\text{Again, } \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial y} = \frac{1}{2} \cdot \frac{2y}{\sqrt{x^2+y^2}} \frac{\partial u}{\partial r} + \frac{1}{\left(1+\left(\frac{y}{x}\right)^2\right)} \left(\frac{1}{x}\right) \frac{\partial u}{\partial \theta} \\
&= \frac{y}{\sqrt{x^2+y^2}} \frac{\partial u}{\partial r} + \frac{x}{x^2+y^2} \frac{\partial u}{\partial \theta} = \frac{y}{r} \frac{\partial u}{\partial r} + \frac{x}{r^2} \frac{\partial u}{\partial \theta} \quad \because r = \sqrt{x^2+y^2} \\
&= \sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \quad \because x = r \cos \theta \text{ and } y = r \sin \theta
\end{aligned}$$

$$\Rightarrow \frac{\partial}{\partial y} (u) = \left(\sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \right) (u)$$

$$\Rightarrow \frac{\partial}{\partial y} = \left(\sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \right)$$

$$\begin{aligned}
\therefore \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) \\
&= \cos^2 \theta \frac{\partial^2 u}{\partial r^2} - \cos \theta \frac{\partial}{\partial r} \left(\frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\cos \theta \frac{\partial u}{\partial r} \right) + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \\
&= \cos^2 \theta \frac{\partial^2 u}{\partial r^2} - \cos \theta \left(\frac{\sin \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} - \frac{\sin \theta}{r^2} \frac{\partial u}{\partial \theta} \right) - \frac{\sin \theta}{r} \left(\cos \theta \frac{\partial^2 u}{\partial r \partial \theta} - \sin \theta \frac{\partial u}{\partial r} \right) + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \\
&= \cos^2 \theta \frac{\partial^2 u}{\partial r^2} - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\sin^2 \theta}{r} \frac{\partial u}{\partial r} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \dots \textcircled{1}
\end{aligned}$$

$$\begin{aligned}
\text{Also, } \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \left(\sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \right) \\
&= \sin^2 \theta \frac{\partial^2 u}{\partial r^2} + \sin \theta \frac{\partial}{\partial r} \left(\frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \right) + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial r} \right) + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \\
&= \sin^2 \theta \frac{\partial^2 u}{\partial r^2} + \sin \theta \left(\frac{\cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} - \frac{\cos \theta}{r^2} \frac{\partial u}{\partial \theta} \right) + \frac{\cos \theta}{r} \left(\sin \theta \frac{\partial^2 u}{\partial r \partial \theta} + \cos \theta \frac{\partial u}{\partial r} \right) + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \\
&= \sin^2 \theta \frac{\partial^2 u}{\partial r^2} + \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} - \frac{\sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\cos^2 \theta}{r} \frac{\partial u}{\partial r} + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \dots \textcircled{2}
\end{aligned}$$

Adding $\textcircled{1}$ and $\textcircled{2}$, we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0 \text{ is the required transformation.}$$

1.9.4 Jacobians: If u and v are functions of two independent variables x and y , then the

determinant $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ is called the Jacobian of u, v with respect to x, y and is denoted by

the symbol $J\left(\frac{u,v}{x,y}\right)$ or $\frac{\partial(u,v)}{\partial(x,y)}$. Similarly If u, v, w are functions of x, y, z , then the Jacobian

of u, v, w with respect to x, y, z is $J\left(\frac{u,v,w}{x,y,z}\right)$ or $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ and is given by: $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$

Chain Rule for Jacobians

If $u = f(r, s)$, $v = g(r, s)$ and $r = \phi(x, y)$, $s = \psi(x, y)$, then $\frac{\partial(u,v)}{\partial(x,y)} = \frac{\partial(u,v)}{\partial(r,s)} \cdot \frac{\partial(r,s)}{\partial(x,y)}$

Example25 If (x, y, z) and (ρ, θ, ϕ) are respectively cartesian and spherical coordinates of a point, show that $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = -\rho^2 \sin \phi$.

Solution: Spherical coordinates are given by:

$$x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi$$

$$\therefore \frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix} = \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \\ \cos \phi & 0 & -\rho \sin \phi \end{vmatrix}$$

Taking out $\rho \sin \phi$ from second column and ρ from third column, we get

$$\begin{aligned} \frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} &= \rho^2 \sin \phi \begin{vmatrix} \sin \phi \cos \theta & -\sin \theta & \cos \phi \cos \theta \\ \sin \phi \sin \theta & \cos \theta & \cos \phi \sin \theta \\ \cos \phi & 0 & -\sin \phi \end{vmatrix} \\ &= \rho^2 \sin \phi [\sin \phi \cos \theta (-\sin \phi \cos \theta) - \sin \phi \sin \theta (\sin \theta \sin \phi) + \\ &\quad \cos \phi (-\sin^2 \theta \cos \phi - \cos^2 \theta \cos \phi)] \\ &= \rho^2 \sin \phi [-\sin^2 \phi \cos^2 \theta - \sin^2 \phi \sin^2 \theta - \sin^2 \theta \cos^2 \phi - \cos^2 \theta \cos^2 \phi] \\ &= \rho^2 \sin \phi [-\sin^2 \phi (\cos^2 \theta + \sin^2 \theta) - \cos^2 \phi (\sin^2 \theta + \cos^2 \theta)] \\ &= \rho^2 \sin \phi [-\sin^2 \phi - \cos^2 \phi] = -\rho^2 \sin \phi \end{aligned}$$

Jacobians to Determine Functional Dependence

Two functions $f(x, y)$ and $g(x, y)$ are called functionally dependent if they are functions of each other. In terms of Jacobians, if $f(x, y)$ and $g(x, y)$ are functionally dependent, then

$$\frac{\partial(f, g)}{\partial(x, y)} = 0$$

Example 26 $u = \sin^{-1} x + \sin^{-1} y$, $v = x\sqrt{1-y^2} + y\sqrt{1-x^2}$, find if u and v are functionally dependent, if so then find the relation between them.

Solution:
$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1}{\sqrt{1-x^2}} & \frac{1}{\sqrt{1-y^2}} \\ \sqrt{1-y^2} + \frac{1}{2} \frac{y(-2x)}{\sqrt{1-x^2}} & \sqrt{1-x^2} + \frac{1}{2} \frac{x(-2y)}{\sqrt{1-y^2}} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{1}{\sqrt{1-x^2}} & \frac{1}{\sqrt{1-y^2}} \\ \sqrt{1-y^2} & \sqrt{1-x^2} \end{vmatrix} + \begin{vmatrix} 0 & 0 \\ \frac{-xy}{\sqrt{1-x^2}} & \frac{-xy}{\sqrt{1-y^2}} \end{vmatrix} = 1 - 1 + 0 = 0$$

$\therefore u$ and v are functionally dependent

$$\begin{aligned} \text{Also } \sin^{-1} v &= \sin^{-1} (x\sqrt{1-y^2} + y\sqrt{1-x^2}) \\ &= \sin^{-1} x + \sin^{-1} y = u \end{aligned}$$

$$\therefore \sin^{-1} v = u \text{ or } v = \sin u$$

Jacobians of Implicit Functions

Let u, v be implicit functions of variables x, y and connected by the relations

$$f_1(u, v, x, y) = 0 \text{ and } f_2(u, v, x, y) = 0, \text{ then } \frac{\partial(u, v)}{\partial(x, y)} = (-1)^2 \frac{\frac{\partial(f_1, f_2)}{\partial(x, y)}}{\frac{\partial(f_1, f_2)}{\partial(u, v)}}$$

Partial Derivatives of Implicit Functions Using Jacobians

- $\frac{\partial u}{\partial x} = -\frac{\frac{\partial(f_1, f_2)}{\partial(x, v)}}{\frac{\partial(f_1, f_2)}{\partial(u, v)}}, \quad \frac{\partial u}{\partial y} = -\frac{\frac{\partial(f_1, f_2)}{\partial(y, v)}}{\frac{\partial(f_1, f_2)}{\partial(u, v)}}$
- $\frac{\partial v}{\partial x} = -\frac{\frac{\partial(f_1, f_2)}{\partial(u, x)}}{\frac{\partial(f_1, f_2)}{\partial(u, v)}}, \quad \frac{\partial v}{\partial y} = -\frac{\frac{\partial(f_1, f_2)}{\partial(u, y)}}{\frac{\partial(f_1, f_2)}{\partial(u, v)}}$

Again, if u, v, w be implicit functions of variables x, y, z and connected by the relations

$$f_1(u, v, w, x, y, z) = 0, f_2(u, v, w, x, y, z) = 0 \text{ and } f_3(u, v, w, x, y, z) = 0$$

$$\text{then } \frac{\partial(u, v, w)}{\partial(x, y, z)} = (-1)^3 \frac{\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)}}{\frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)}}$$

Note: If $J = \frac{\partial(u,v,w)}{\partial(x,y,z)}$, then $\frac{1}{J} = \frac{\partial(x,y,z)}{\partial(u,v,w)}$

Result: If u and v are functions of two independent variables x and y , such that

$$J = \frac{\partial(u,v)}{\partial(x,y)}, \quad J' = \frac{\partial(x,y)}{\partial(u,v)}, \quad \text{then } JJ' = 1, \text{ where } J' = \frac{1}{J}$$

Proof: Let $u = u(x, y) \quad \dots (1)$

$$v = v(x, y) \quad \dots (2)$$

Differentiating (1) with respect to u and v

$$\frac{\partial u}{\partial u} = 1 = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial u} \quad \text{and} \quad \frac{\partial u}{\partial v} = 0 = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial v}$$

Differentiating (2) with respect to u and v

$$\frac{\partial v}{\partial u} = 0 = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial u} \quad \text{and} \quad \frac{\partial v}{\partial v} = 1 = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$\text{Now } JJ' = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)}$$

$$\begin{aligned} &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \cdot \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial u} & \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial v} \\ \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial u} & \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial v} \end{vmatrix} \\ &= \begin{vmatrix} \frac{\partial u}{\partial u} & \frac{\partial u}{\partial v} \\ \frac{\partial v}{\partial u} & \frac{\partial v}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \end{aligned}$$

Hence Proved

Example27 For the transformation $x = e^u \cos v$, $y = e^u \sin v$, show that $\frac{\partial(u,v)}{\partial(x,y)} = e^{-2u}$

$$\text{Solution: Let } J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} e^u \cos v & -e^u \sin v \\ e^u \sin v & e^u \cos v \end{vmatrix}$$

$$= e^u \cdot e^u \begin{vmatrix} \cos v & -\sin v \\ \sin v & \cos v \end{vmatrix} = e^{2u}(1)$$

$$\therefore \frac{\partial(u,v)}{\partial(x,y)} = e^{-2u}$$

Example28 If $x^2 + y^2 + u^2 - v^2 = 0$ and $uv + xy = 0$, show that $\frac{\partial(u,v)}{\partial(x,y)} = \frac{x^2 - y^2}{u^2 + y^2}$

Solution: We have $x^2 + y^2 + u^2 - v^2 = 0$ and $uv + xy = 0$

$$\Rightarrow f_1(u, v, x, y) = 0 \text{ and } f_2(u, v, x, y) = 0$$

$$\therefore \frac{\partial(u, v)}{\partial(x, y)} = (-1)^2 \frac{\frac{\partial(f_1, f_2)}{\partial(x, y)}}{\frac{\partial(f_1, f_2)}{\partial(u, v)}}$$

$$\begin{aligned} \Rightarrow \frac{\partial(u, v)}{\partial(x, y)} &= \frac{\begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{vmatrix}}{\begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \end{vmatrix}} y^2 \\ &= \frac{\begin{vmatrix} 2x & 2y \\ y & x \end{vmatrix}}{\begin{vmatrix} 2u & 2v \\ v & u \end{vmatrix}} = \frac{2x^2 - 2y^2}{2u^2 + 2v^2} = \frac{x^2 - y^2}{u^2 + v^2} \end{aligned}$$

Example 29 If $u = x + y^2$, $v = y + z^2$ and $w = z + x^2$, show that $\frac{\partial x}{\partial u} = \frac{1}{1 + 8xyz}$

Solution: Here $f_1 = u - x - y^2 = 0$, $f_2 = v - y - z^2 = 0$, $f_3 = w - z - x^2 = 0$

$$\begin{aligned} \therefore \frac{\partial x}{\partial u} &= - \frac{\frac{\partial(f_1, f_2, f_3)}{\partial(u, v, z)}}{\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)}} \\ &= - \frac{\begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial u} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{vmatrix}}{\begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{vmatrix}} \\ &= - \frac{\begin{vmatrix} 1 & -2y & 0 \\ 0 & -1 & -2z \\ 0 & 0 & -1 \end{vmatrix}}{\begin{vmatrix} -1 & -2y & 0 \\ 0 & -1 & -2z \\ -2x & 0 & -1 \end{vmatrix}} \\ \Rightarrow \frac{\partial x}{\partial u} &= - \frac{1(1)}{-1(1) - 2x(4yz)} = \frac{1}{1 + 8xyz} \end{aligned}$$